Reverse of differentiation

1

a $4x^3$

b $x^4 + C$

2

a $6x^5$

b $4x^6 + C$

3

a $5x^4 + 4x^3$

b $7x^5 + 7x^4 + C$

4

a $-4x^{-5}$

b $-4x^{-4} + C$

5

a $8x^7 + 6x^5$

b $\frac{1}{2}x^8 + \frac{1}{2}x^6 + C$

6

a $\frac{6}{5}\sqrt[5]{x}$

b $5\sqrt[5]{x^6} + C$

Anti-differentiation

7

a $f(x) = 2x + C$ **b** Linear **c** $f(x) = mx + C$

8

a $f(x) = -4x^2 + C$

b Quadratic

c $f(x) = \frac{a}{2}x^2 + C$

9

a $f(x) = 2x^2 + 3x + C$

b Quadratic

c $f(x) = \frac{a}{2}x^2 + bx + C$

10

a $f(x) = \frac{7x^3}{3} + C$

b Cubic

c $f(x) = \frac{a}{3}x^3 + C$

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a $y = 9x + C$

b $y = 4x^2 + C$

c $y = 5x^2 + 7x + C$

d $y = 3x^3 + C$

e $y = 3x^3 + 2x^2 - 6x + C$

f $y = \frac{x^6}{6} + C$

g $y = 3x^5 + 4x^4 + C$

h $y = \frac{x^7}{28} + \frac{x^3}{9} + C$

i $y = -\frac{x^{-5}}{5} + C$

j $y = -\frac{3}{x^5} + C$

k $y = -\frac{2}{x^5} + \frac{3}{x^3} + C$

l $y = \frac{20x^{\frac{7}{5}}}{7} + \frac{21x^{\frac{11}{7}}}{11} + C$

m $y = \frac{7x^{\frac{4}{7}}}{4} + \frac{5x^{\frac{3}{5}}}{3} + C$

n $y = 2x^4 + \frac{9x^{\frac{8}{3}}}{8} - 3x + C$

o $y = x - 2x^{-2} + C$

p $y = 2x^5 - \frac{9x^4}{4} + C$

q $y = 5x^3 - 13x^2 + 8x + C$

r $y = \frac{x^3}{3} + 5x^2 + 24x + C$

s $y = \frac{2x^{\frac{3}{2}}}{3} + C$

t $y = 12\sqrt{x^3} + C$

u $y = \frac{2x^{\frac{5}{2}}}{5} + C$

v $y = 12\sqrt{x} + C$

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Indefinite integrals

12

a $4x + C$

b $\frac{x^4}{4} + C$

c $x^4 + C$

d $\frac{x^3}{12} + C$

e $\frac{x^6}{6} + C$

f $\frac{x^3}{3} + 2x^2 + C$

g $5t - \frac{3t^2}{2} - \frac{4t^3}{3} + C$

h $x^4 + x^3 + C$

i $x^3 + 3x^2 + 3x + C$

j $\frac{3x^2}{2} + x^3 + \frac{x^4}{4} + C$

k $\frac{x^3}{15} + \frac{x^4}{16} + 3x + C$

l $-\frac{x^{-4}}{4} + C$

m $-\frac{3}{x} - 4x^2 + C$

n $-\frac{5}{x} + C$

o $\frac{4x^{\frac{5}{4}}}{5} + C$

p $\frac{5x^{\frac{9}{5}}}{9} + C$

q $\frac{20x^{\frac{7}{4}}}{7} + C$

r $\frac{2x^{\frac{3}{2}}}{3} + C$

s $\frac{5x^{\frac{9}{5}}}{9} + 3x^{\frac{5}{3}} + C$

t $\frac{3}{2}x^{\frac{10}{3}} + \frac{4}{15}x^{\frac{9}{4}} + C$

u $3x^{\frac{5}{3}} + 2x^{\frac{3}{2}} + 6x + C$

v $2x^{\frac{1}{2}} + C$

w $x^3 + 8x^2 - 12x + C$

x $7x^4 + 19x^3 + 7x^2 + C$

y $x^7 + 3x^3 + C$

z $\frac{7x^4}{4} - \frac{14x^3}{3} + \frac{7x^2}{2} + C$

13

a $\frac{49x^8}{8} + \frac{42x^5}{5} + \frac{9x^2}{2} + C$

b $\frac{x^2}{2} - \frac{8x^{\frac{3}{2}}}{3} + 4x + C$

c $\frac{x^2}{2} + 20x^{\frac{1}{2}} - 25x^{-1} + C$

d $2x^{\frac{7}{2}} + 2x^{\frac{5}{2}} + C$

e $\frac{x^4}{12} + \frac{3}{2x^2} + C$

f $x^5 - 2x + C$

g $4x^2 - 3x^{-3} + C$

h $\frac{2x^{\frac{7}{2}}}{7} + 10x^{\frac{1}{2}} + C$

i $\frac{2y^{\frac{5}{2}}}{5} + C$

j $-2y^{\frac{-1}{2}} + C$

k $\frac{ay^4}{4} + C$

l $\frac{2t^{\frac{3}{2}}}{3n} + C$

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Applications

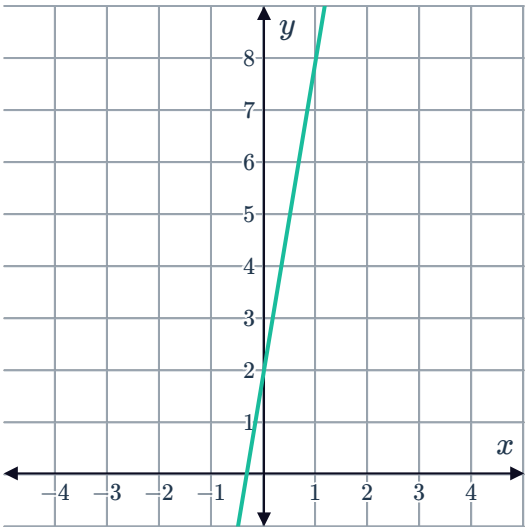
14

a

i Linear

ii $f(x) = 6x + C$

iii Any straight line with gradient of 6:

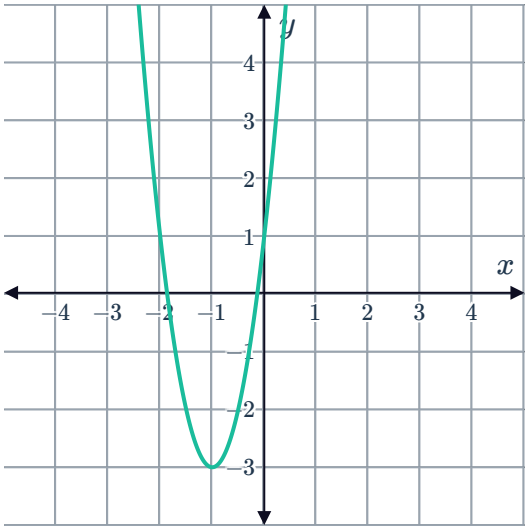


b

i Quadratic

ii $f(x) = 4x^2 + 8x + C$

iii Any parabola of the form $f'(x) = 4x^2 + 8x + C$ with axis of symmetry $x = -1$:



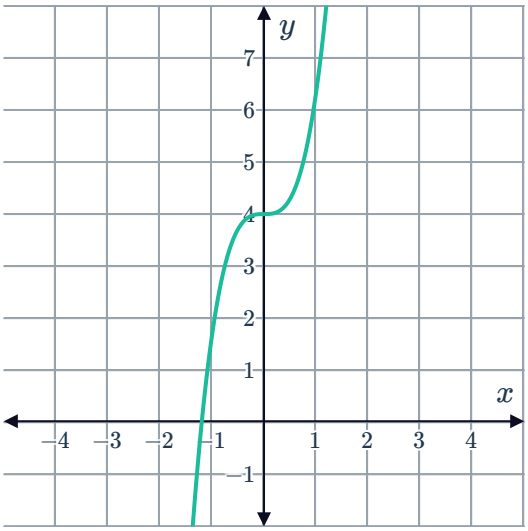
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c

i Cubic

ii $f(x) = \frac{7x^3}{3} + C$

iii Any cubic of the form $\frac{7x^3}{3} + C$:

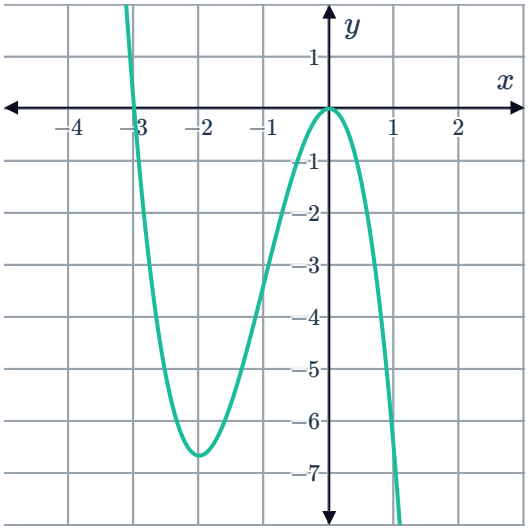


d

i Cubic

ii $f(x) = -\frac{5x^3}{3} - 5x^2 + C$

iii Any cubic of the form $f(x) = -\frac{5x^3}{3} - 5x^2 + C$:

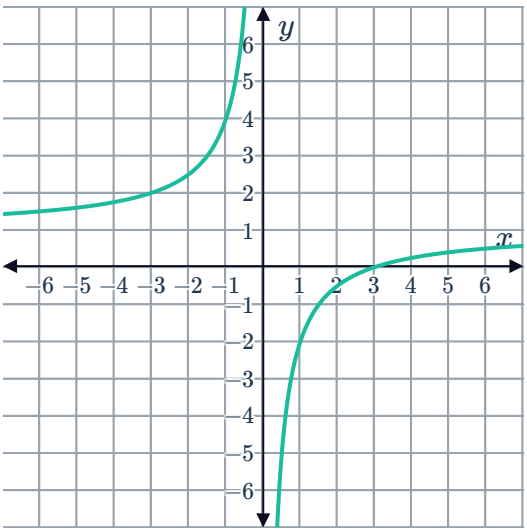


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15

a $f(x) = -\dfrac{3}{x} + C$

b Any graph of a rational function of the form $f(x) = -\dfrac{3}{x} + C$ with vertical asymptote at $x = 0$ and horizontal asymptote at $y = C$:

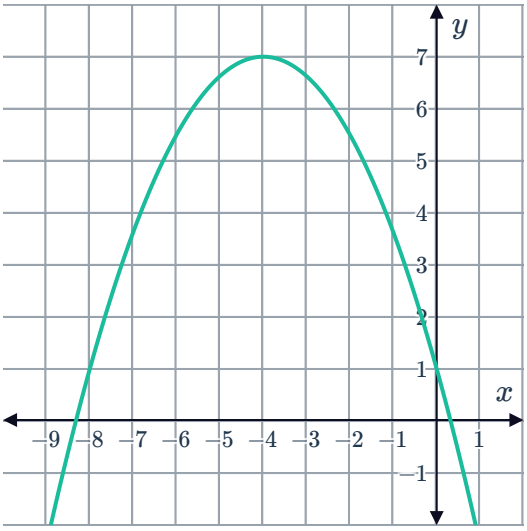


16

a $f'(x) = -\dfrac{3x}{4} - 3$

b $f(x) = -\dfrac{3x^2}{8} - 3x + C$

c Any parabola of the form $f(x) = -\dfrac{3x^2}{8} - 3x + C$ with axis of symmetry $x = -4$:



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Find value of C

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a $y = 2x^2 + 7x + C$

b $y = 2x^2 + 7x + 2$

18

a $y = 3x^3 - 5x^2 + 2x + C$

b $y = 3x^3 - 5x^2 + 2x + 5$

19

a $y = 5x^3 + 7x + C$

b $y = 5x^3 + 7x + 5$

20

a $y = 2x^5 + 5x^4 + 2x^3 + 3x^2 + 9x + C$

b $y = 2x^5 + 5x^4 + 2x^3 + 3x^2 + 9x + 2$

21

a $y = \frac{27x^{\frac{5}{3}}}{5} + C$

b $y = \frac{27x^{\frac{5}{3}}}{5} + 5$

22 $p = 3t^2 - 5t + 3$

23 $y = 3x + 2$

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